CLICK-WHEEL MUSIC PLAYER · 2004–2009

Rotunda Psalter

A pocket psalter that scrolls like a music library: split-screen menus, brushed title bars, a flat Cover Flow — and the day as an album of four offices.

The X3 has the same four controls as the click-wheel player — scroll, select, back, play — so Rotunda borrows its whole grammar. Today's Office is an album with four tracks at 8:30, 12:00, 4:30 and 8:00; the clock selects the track that's due. Book clubs are playlists. Revision B drops the cover reflections from the lock screen and library: they cost rendering time and SD space and added little.

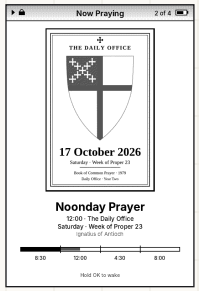
DESIGN PRINCIPLES

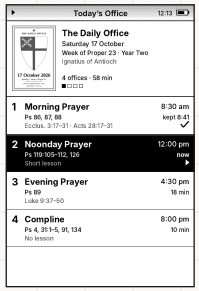
- Selection is always solid black: moving it is a cheap 1-bit partial refresh.
- Grey appears only in covers and the title-bar dither, which change on full refresh.
- Every list is one column, so ◀/▶ alone reaches everything.

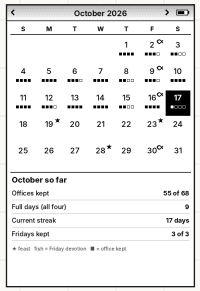
NOSTALGIC TOUCHES

Brushed-metal title bar · Split-screen menu with art pane · Flat Cover Flow · Album & tracks for the day · Playlists for clubs · 'Shuffle Psalms' · Scrubber & '1 of 3'

ALSO IN THIS SPEC









HOME SCREEN · ACTUAL SIZE ×0.9



PROJECT	PANEL	REV	FOR	SHEET
Daily Office · Xteink X3 custom firmware UI	528 × 792 · 3.7″ · 4 inks	A · 17 Oct 2026	Robert E. Pinard	1 of 10



SCREEN 01 OF 08

Lock / Sleep screen

Shown while the device sleeps. Rendered once, with the full grey waveform, then the panel is unpowered — an e-ink image costs nothing to hold.

ANNOTATIONS

1. Hold glyph replaces the play arrow while asleep.
2. Today's EPUB cover, pre-dithered once and cached — no reflection, no extra rendering.
3. Office due at the moment of sleep, from the clock.
4. Commemoration in small grey type.
5. Day scrubber: four segments labelled with the office hours.

PIXEL DETAIL · 1:1 AT ×4

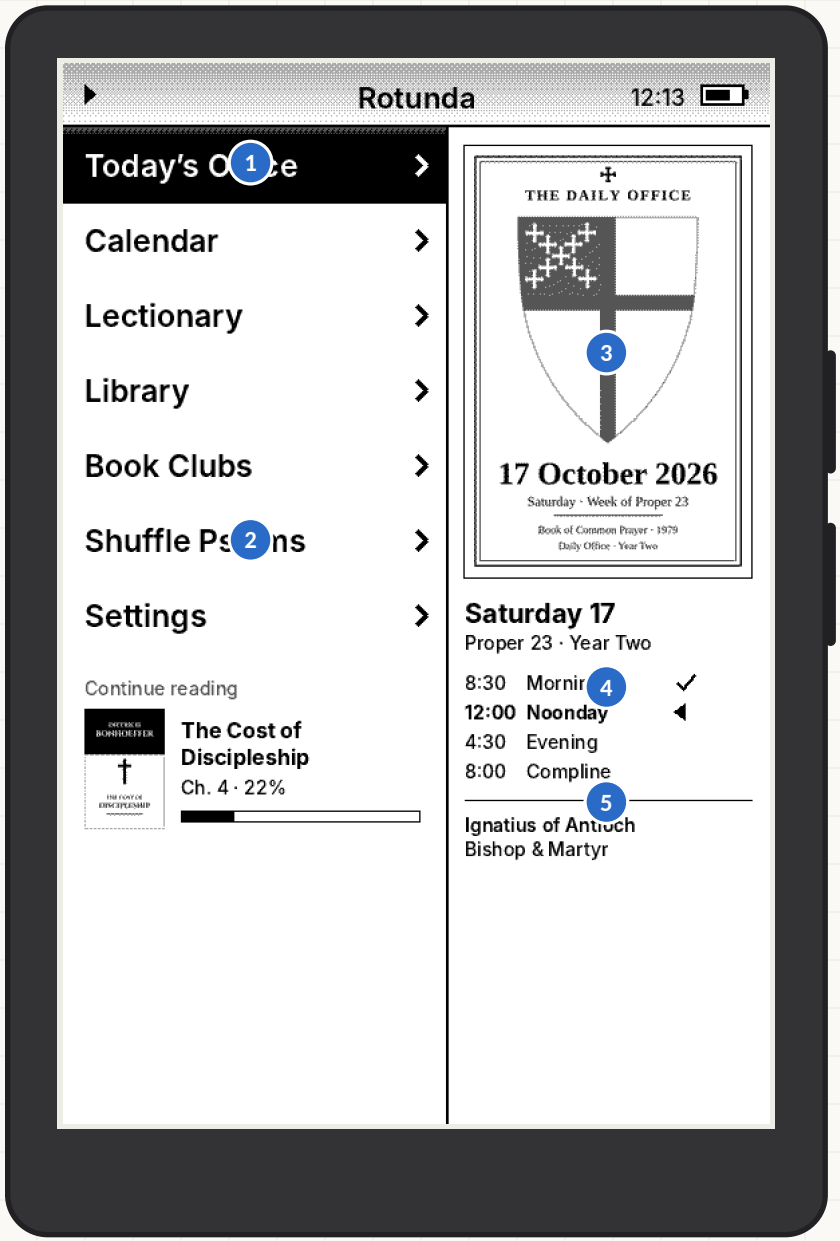


px 198–330, 510–609 · inks 00 55 AA FF only

ENGINEERING NOTE

Refresh: one full 4-grey refresh at sleep. Wake: OK (or any key).

PROJECT	PANEL	REV	FOR	SHEET
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SCREEN 02 OF 08

Home

The launch point after wake. Shows the day at a glance and opens today's office in one press.

ANNOTATIONS

1. Five sections: Today's Office, Calendar, Lectionary, Library, Book Clubs.
2. 'Shuffle Psalms' opens a random psalm.
3. Art pane follows the highlighted row.
4. Today at a glance: hours, ✓ kept, ◀ due now.
5. Commemoration under a hairline rule.

PIXEL DETAIL · 1:1 AT ×4

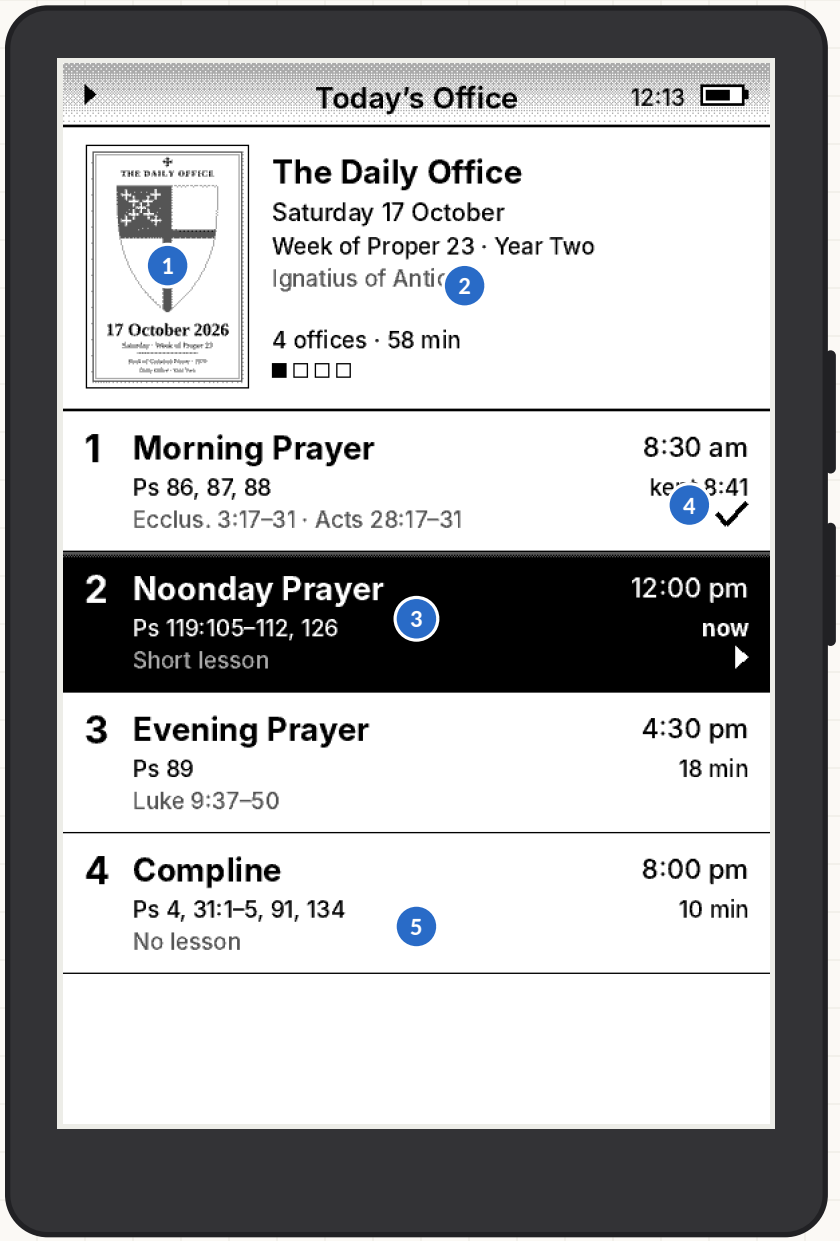


px 74–206, 24–123 · inks 00 55 AA FF only

ENGINEERING NOTE

Refresh: full on entry; focus moves with a fast 1-bit partial update of the two affected regions.

PROJECT	PANEL	REV	FOR	SHEET
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SCREEN 03 OF 08

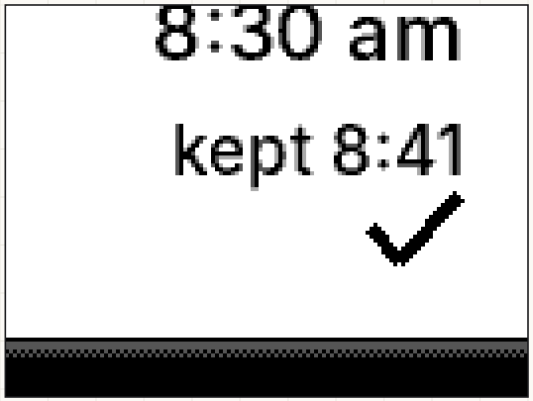
Today's Office

All four offices of the day on one screen, with their hours (8:30 · 12:00 · 4:30 · 8:00), psalms, lessons and whether each has been kept. The clock picks the current office and puts focus on it, so a single OK opens the right chapter.

ANNOTATIONS

1. The day is an 'album'; its four offices are 'tracks'.
2. Commemoration under the date.
3. Clock-selected office: highlighted and focused on entry — OK plays it.
4. Kept offices show the time they were finished.
5. Hours fixed at 8:30 · 12:00 · 4:30 · 8:00.

PIXEL DETAIL · 1:1 AT ×4

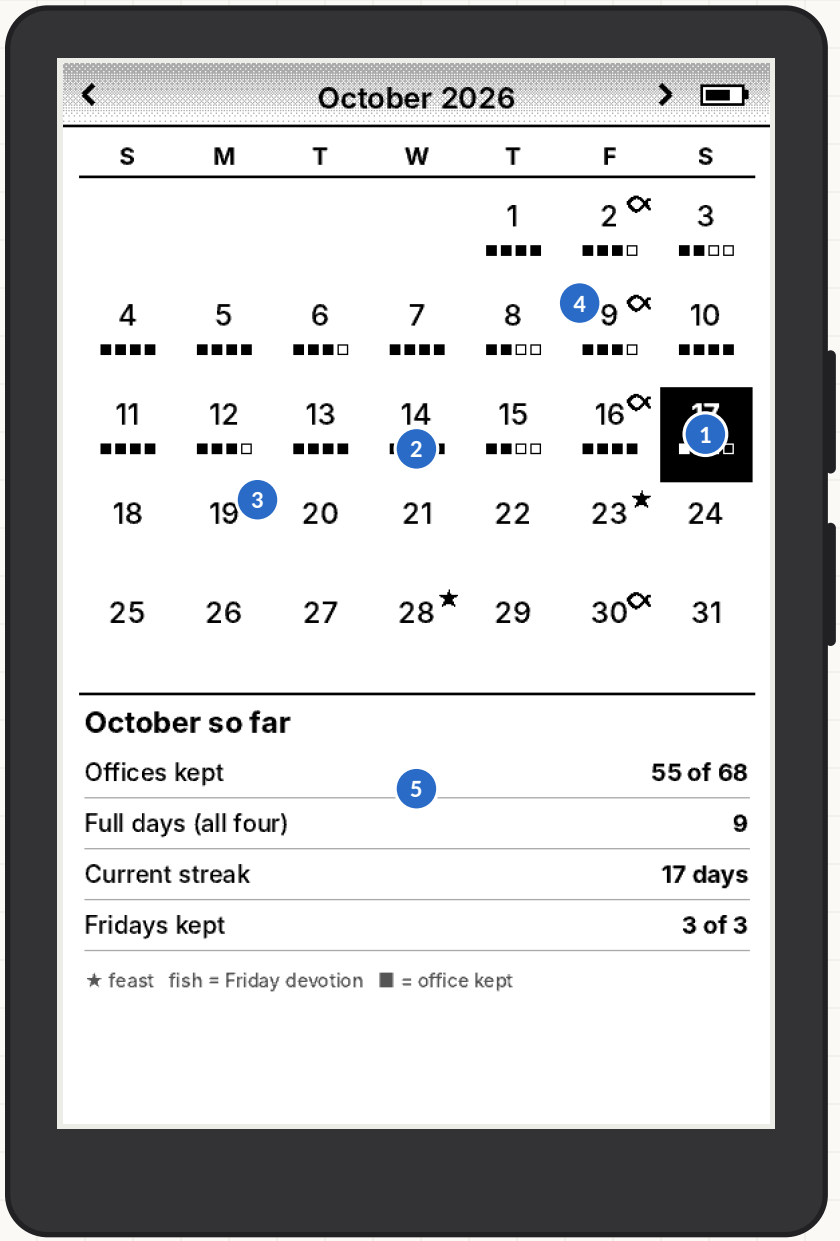


px 396–528, 280–379 · inks 00 55 AA FF only

ENGINEERING NOTE

Current office from the real-time clock: Morning 4:00 am–11:59 · Noonday 12:00–4:29 · Evening 4:30–7:59 · Compline 8:00 pm–3:59 am.

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SCREEN 04 OF 08

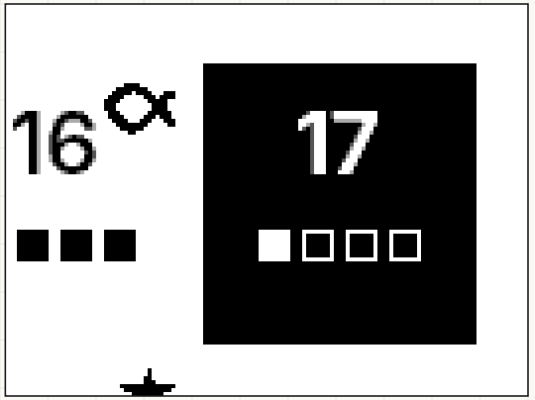
Monthly calendar

The month at a glance: how many of the four offices were kept each day, with feasts and Friday devotion marked. A day can be opened to see what was kept.

ANNOTATIONS

1. Today inverted; ◀/▶ walk days, hold for weeks.
2. Four squares per day: filled = office kept.
3. Star in the corner = red-letter feast.
4. Fish = Friday devotion.
5. Month totals: offices, full days, streak, Fridays.

PIXEL DETAIL · 1:1 AT ×4

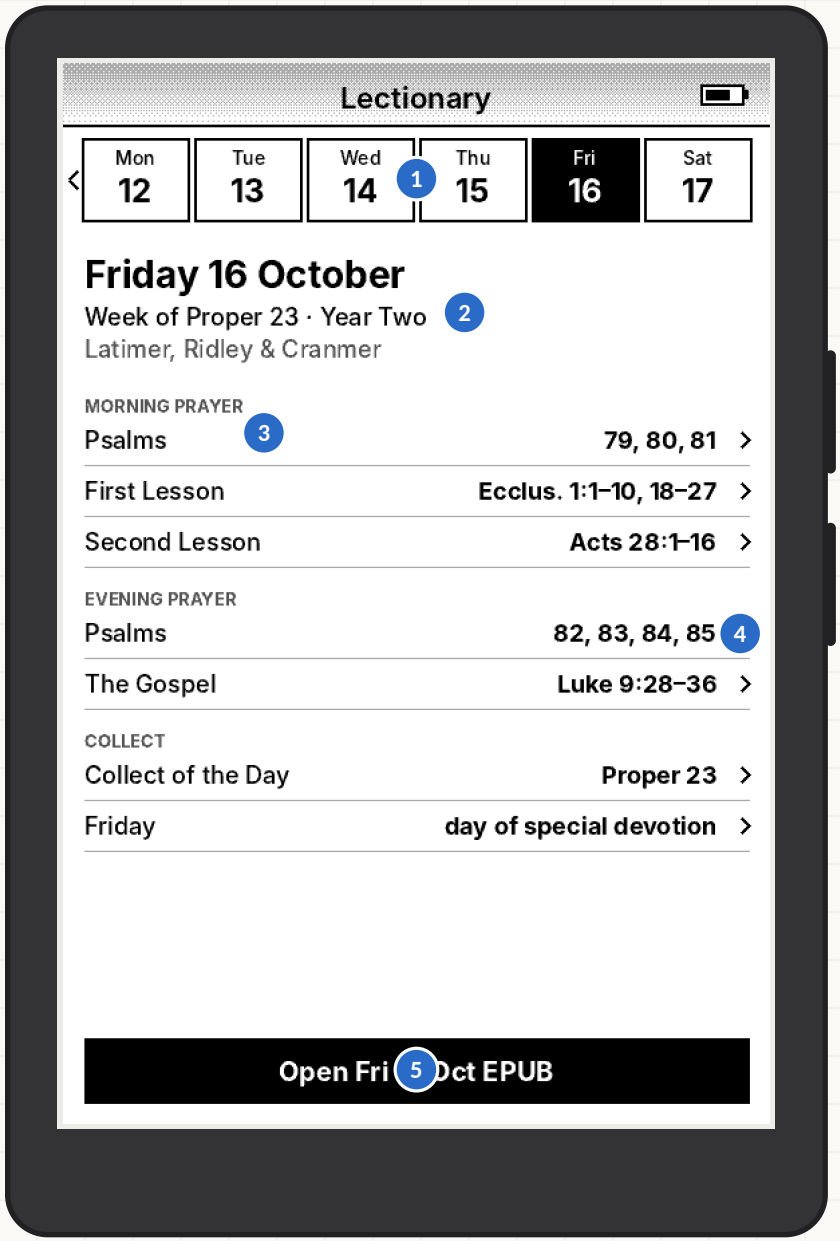


px 396–528, 227–326 · inks 00 55 AA FF only

ENGINEERING NOTE

Data: /office/said.csv (date, office, time) written when an office's last chapter is read; feast rank and fast days from /office/calendar.csv.

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SCREEN 05 OF 08

Lectionary archive

The lectionary archive: step back through past days to see what was appointed — the Proper, psalms by the 30-day cycle, the three lessons, the collect and any commemoration — and reopen that day's EPUB if it is still on the card.

ANNOTATIONS

1. Day strip: ◀ steps back through past days (any date with an index row).
2. Selected day: Proper, year and commemoration (here Latimer, Ridley & Cranmer).
3. Every reading appointed that day, with the 30-day psalms.
4. Each row opens straight to that reading if the EPUB is on the card.
5. Black action bar: reopen that day's whole book.

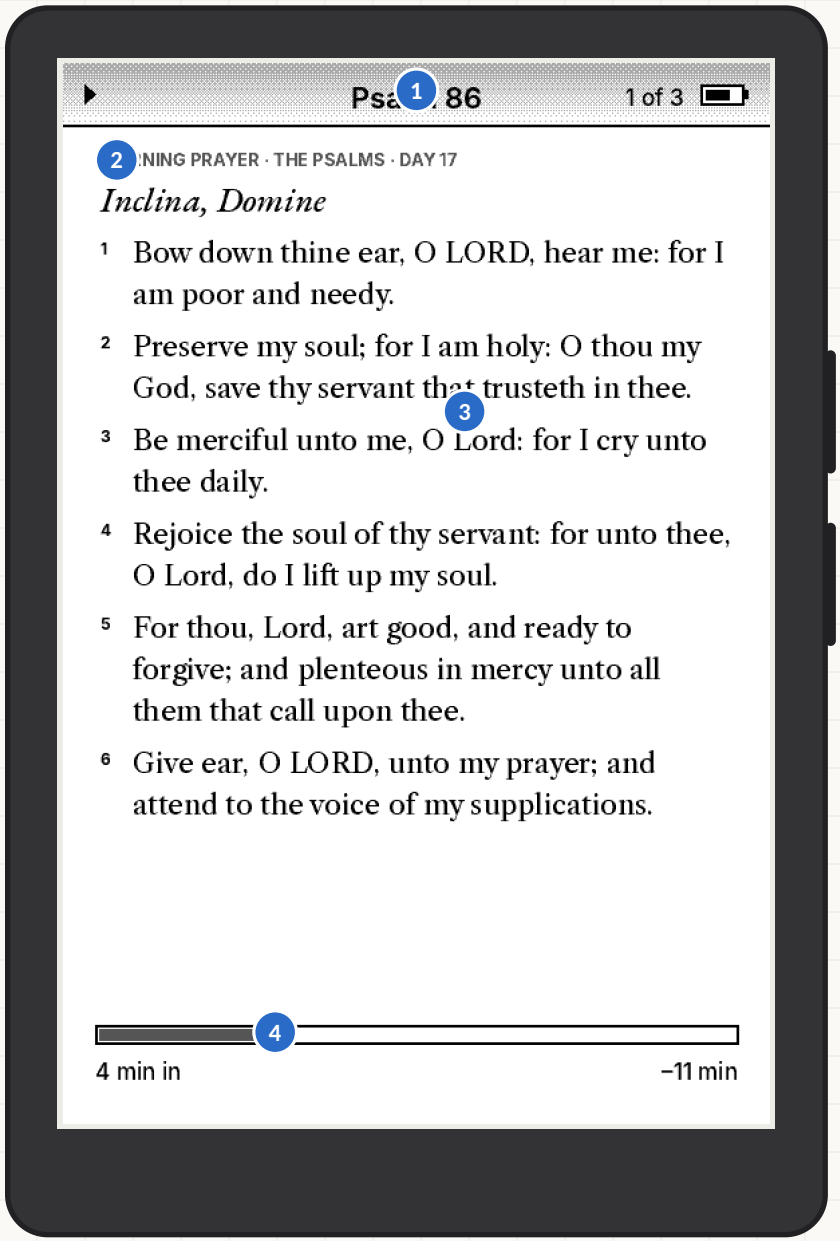
PIXEL DETAIL · 1:1 AT ×4



px 198–330, 693–792 · inks 00 55 AA FF only

ENGINEERING NOTE

Data: /office/YYYY-MM.csv written by the EPUB generator, so past days are browsable even after their EPUBs are deleted.



SCREEN 06 OF 08

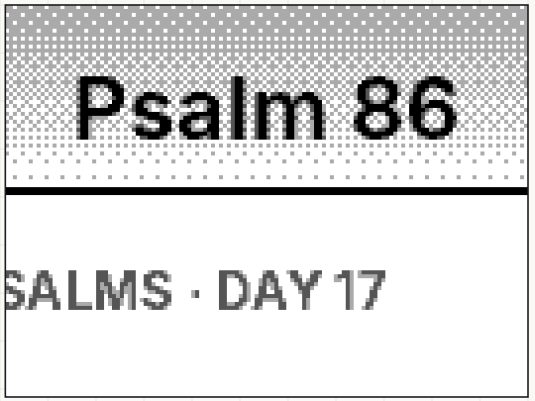
Reader

Chrome around the reading engine. The theme styles headers, footers and soft keys only; body text is laid out by the reader from the EPUB.

ANNOTATIONS

1. Title bar: psalm and ‘1 of 3’ psalms appointed.
2. Running head from the EPUB.
3. Body text belongs to the reader.
4. Scrubber shows minutes left in this office.

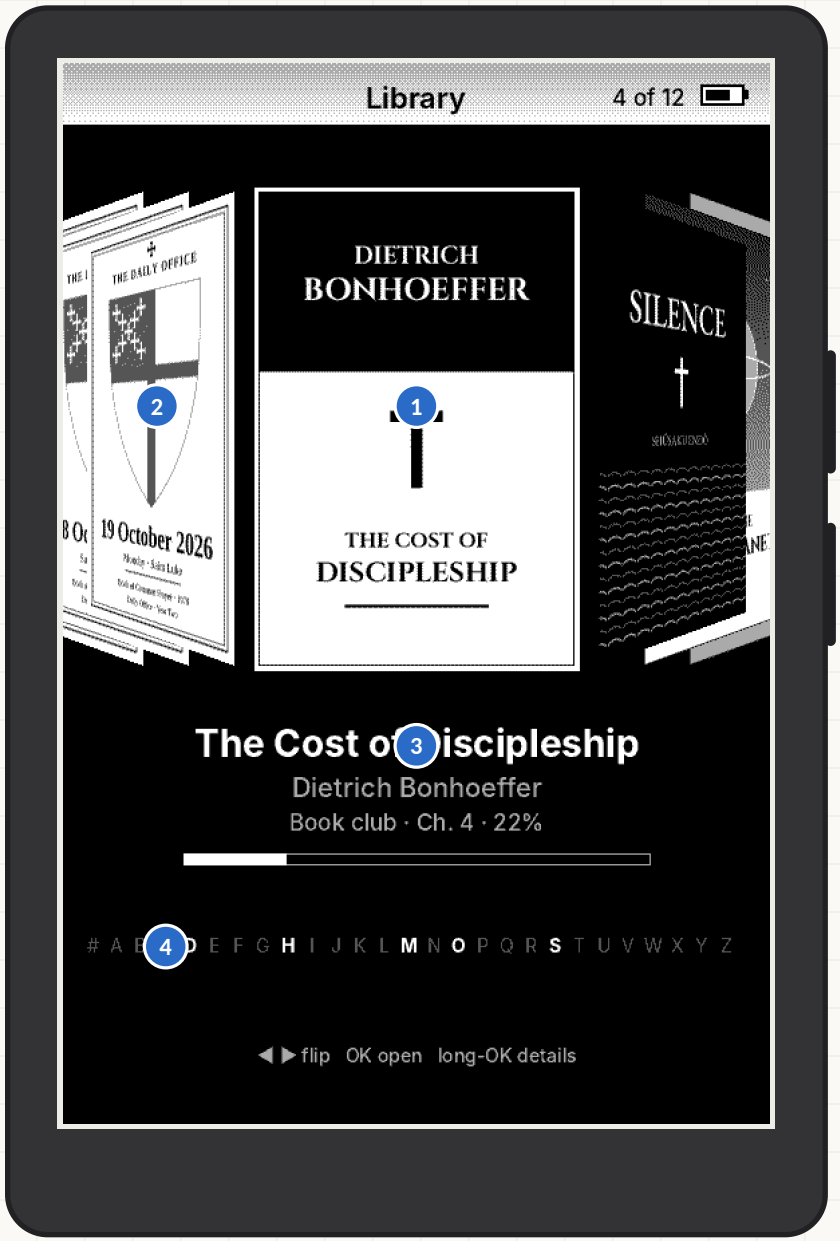
PIXEL DETAIL · 1:1 AT ×4



px 198–330, 0–99 · inks 00 55 AA FF only

ENGINEERING NOTE

Refresh: 1-bit partial per page, full refresh every N pages (user setting) to clear ghosting.



SCREEN 07 OF 08

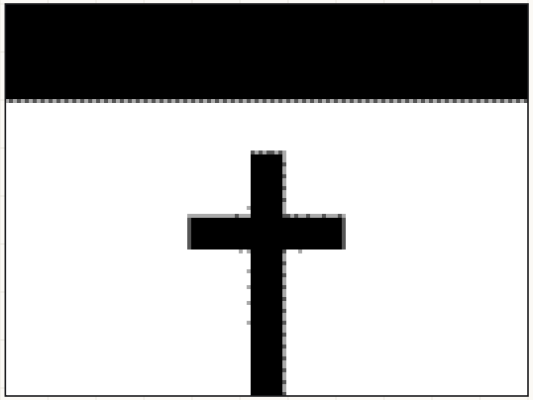
Library

Every book on the SD card, with covers. Covers are scaled and dithered once at import, cached under .crosspoint/, and never resampled at runtime.

ANNOTATIONS

1. Centre cover: flat 4-grey thumbnail.
2. Side covers skewed once at import and cached; no reflections to draw or store.
3. Title, author, shelf and progress.
4. Alphabet jump strip.

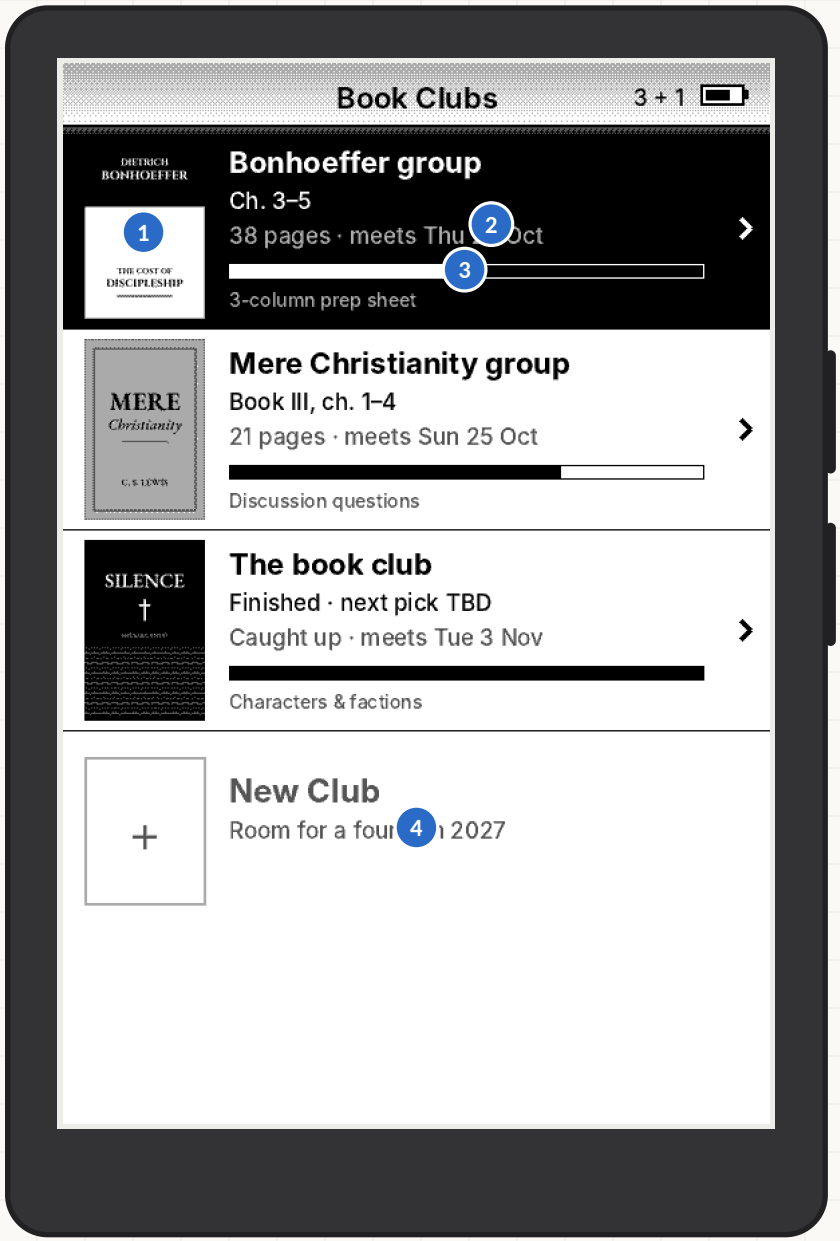
PIXEL DETAIL · 1:1 AT ×4



px 198–330, 206–305 · inks 00 55 AA FF only

ENGINEERING NOTE

Refresh: grey refresh only when a page of covers first appears; focus moves are 1-bit.



SCREEN 08 OF 08

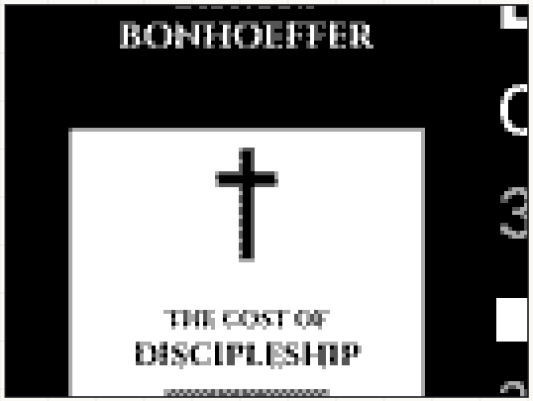
Book clubs

Up to four book clubs, each with its current book, this week's assignment, pages left before the meeting and a link to your prep notes. Assignments map to the EPUB's table of contents, so 'Ch. 3–5' becomes a page count the reader can track.

ANNOTATIONS

1. Each club is a 'playlist': cover art, name, this week's assignment.
2. Pages left and meeting date (sample dates).
3. Assignment progress bar.
4. Fourth slot for next year.

PIXEL DETAIL · 1:1 AT ×4



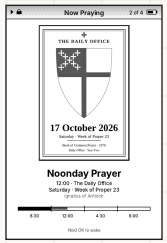
px 0–132, 76–175 · inks 00 55 AA FF only

ENGINEERING NOTE

Data: /clubs.csv on SD (club, book file, start & end chapter, meeting date, notes file). Meeting dates in these renders are sample data.

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ALL 8 SCREENS



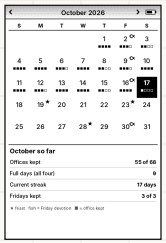
01 LOCK



02 HOME



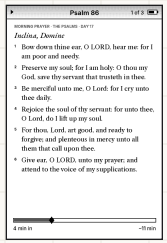
03 TODAY



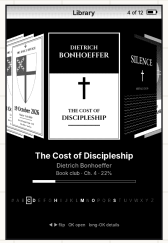
04 CALENDAR



05 LECTIONARY



06 READER



07 LIBRARY



08 CLUBS

THE FOUR INKS

- **Black #000000** — Text, focus, rules
- **Dark grey #555555** — Secondary text, shading
- **Light grey #AAAAAA** — Bands, outlines, dither
- **Paper #FFFFFF** — Grounds

TYPOGRAPHY

Chrome
Inter 500–700, baked to 1-bit bitmaps at 13–32 px

Reading
Libre Caslon Text 21 px (user-selectable)

Accent
EB Garamond Italic for Latin incipits

BUTTON MAP

◀ / ▶	Scroll the list, like turning the wheel
OK	Centre button: select / open
BACK	MENU: up one level; hold = Home
Tilt	Gyro page-turn in reader only

RESOURCE BUDGET (ESP32-C3, ~380 KB USABLE RAM)

Framebuffer	528×792 @ 1 bpp	52 KB
Grey pass scratch	tiled bands	~8 KB
Sleep image	2 bpp BMP on SD	103 KB
Cover thumbs	per book, cached	4–26 KB
Chrome fonts	1-bit bitmaps, flash	≤ 60 KB
Icons & tiles	1-bit, flash	≤ 24 KB
Day index	CSV per month on SD	~3 KB

Figures are design targets, not measurements. Nothing on any screen needs RAM beyond one framebuffer plus the existing grey-band scratch; images stream from SD.

BUILD NOTES

- Fork CrossPoint and theme its activity screens; reading, caching and EPUB layout stay untouched.
- Have the EPUB generator also write /office/YYYY-MM.csv: date, Proper, year, psalms, lessons, feast — the picker reads this instead of parsing EPUBs.
- Chapter anchors per office (#mp, #noon, #ep, #compline) let every 'open office' action jump straight in.
- Log a line to /office/said.csv when an office chapter is finished; streaks, pips and towers all read it.
- Office hours 8:30 · 12:00 · 4:30 · 8:00; the RTC picks the current office (Noonday from 12:00 to 4:29).
- Feast rank, fast days and commemorations come from /office/calendar.csv (one row per day), so the calendar can mark them without parsing books.
- Book clubs read /clubs.csv; pages-left comes from the chapter range in each EPUB's table of contents.
- No animation anywhere. Anything that 'blinked' in the original is drawn static.